

AE-588: Irradiation of [Ru(bpy)3] after irradi of 300 μM Ru(bpy)

Date: 2025-10-13

Tags: O2 [Ru(bpy)3]Cl2*6 H2O AE HTE
Irradiation O2 sensor manual test
Autocatalytic deactivation deactivation

Category: HTE

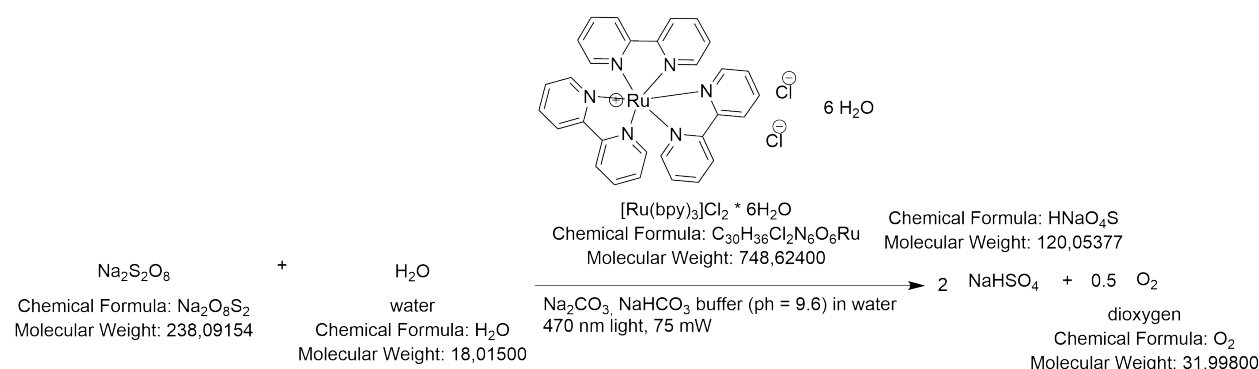
Status: Analytics to be completed

Created by: Alexander Eith

Objective

Reproduction of AE-579 to test reproducibility

Reaction scheme/sample structure



ChemDraw File (linked): [AE-455.cdxml](#)

Literature/reference experiments

Literature	https://doi.org/10.1021/acscatal.6b02595
Reproduction	HTE - AE-579: Irradiation of [Ru(bpy)3] after irradi of 300 μM Ru(bpy)
Similar experiments	HTE - AE-566: Irradiation of [Ru(bpy)3] after irradi of 300 μM Ru(bpy) I

Reagents

Name	CAS Number / Experiment Number	Inventory number	Amount [mmol]	Equivalents	Mass _{exp} [mg]	Used concentration	Molar mass [g/mol]	Density (g/ml)	Volume [ml]	Obtained Concentration in reaction vial [mM]
Na2S2O8	HTE - AE-589: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O (Na2S2O8)	/	/	/	/	60 mM	238.09	/	0.500	6 mM
NaHCO3	HTE - AE-589: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O (NaHCO3)	/	/	/	/	0.9 M	84.006	/	0.381	0.0708 M

Na2CO3	HTE - AE-589: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O (Na2CO3)	/	/	/	/	0.8 M	105.988	/	0.196	0.0292 M
[Ru(bpy)3]Cl2 * 6 H2O	HTE - AE-589: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O (Ru)	/	/	/	/	/	/	/	0.75 + 7.5 µL	300 + 10 µM
Milli-Q water	/	/	/	/	/	/	/	/	3.173	/

Irradiation Parameters

Power measurement was performed in experiment [HTE - AE-579: Irradiation of \[Ru\(bpy\)3\] after irrad of 300 µM Ru\(bpy\)](#)

	Name
Used Set-up	Equipment - Advanced irradiation setup V1.0 I
Irradiation setup	Equipment - Irradiation setup, 2

	Light Source Name	Used power source	Wavelength [nm]	Power Setting [mW]	Analogue Setting [-]
First light source	Light Source - UHP LED 470 nm	Power Sources - BLS-13000-1E	470	75	0.28

Used beam combiner [Name or None]	none
Irradiation distance [cm]	6.5
Thermostat temperature [°C]	25
Stirring speed [rpm]	300
Start time [s], relative to start of log	specified in procedure
End time [s], relative to start of log	specified in procedure

O2/H2 sensor equipment

	Equipment	Used Protocol
Used FireSting	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel (Firesting 2)	Protocol - Degassing and O2 Measurement with sensor cap with air stable substances
Used O2 Sensor	Equipment - Robust probe for liquid O2 measurment	/
Used H2 Sensor	/	/

Procedure/observations

Date	Time	Step	Observations	Pictures
13.10	09:00	In a 10 mL vial water , NaHCO ₃ , Na ₂ CO ₃ , Ox and Ru (750 µL) where added using a 1 mL Eppendorf pipette	AE-588-1	/
	09:02	AE-588-1 was vortexed (Equipment - VWR® VV3, Vortex Mixer , stage 4/6) for apporx. 10 s	yellow-orange solution	dissolved.jpg
	09:03	From AE-588-1 an aliquote (0.4 mL) was taken	AE-588-300-1	/
	09:10	In a 5 mL GL14 PTFE Schlenk flask AE-588-1 (2.0 mL) was added using a 1 mL Eppendorf pipette. The flasks was equipped with a Equipment - Robust probe for liquid O2 measurment and placed in the Equipment - Advanced irradiation setup V1.0 I	AE-588-300	/
	09:15	The degassing of AE-588-300 was started according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	no change	/
	09:23	The oxygen logging was started	/	/
	09:31	The degassing was stopped	/	/

	09:32	The logging was stopped	/	/
	09:32	The oxygen logging was started	/	/
	09:40	The irradiation was started after 7:30	/	irrad 1 bad quality.jpg
	09:50	To AE-588-300-1 water (0.8 mL) was added using a 1 mL Eppendorf pipette	/	/
	09:57	An UV/Vis of AE-588-300-1 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I)	AE-588-300-1	/
	10:02	The irradiation was stopped after 30:30	/	/
	10:11	The logging was stopped	darker, more greenish-brownish	after irrad 1 bad quality.jpg
	10:15	0.23 mL of AE-588-300 were transferred into a 5 mL vial and water (0.46 mL) was added using a 1 mL Eppendorf pipette	AE-588-300-2	/
		1.5 mL of AE-588-300 were transferred into a 5 mL vial	AE-588-10	/
	10:18	To AE-588-10 Ru (7.5 µL) was added	no change	with new Ru.jpg
	10:19	From AE-588-10 0.4 mL were transferred into another vial and 0.8 mL water was added	AE-588-10-1	/
	10:21	From AE-588-10 1.0 mL were transferred into the irradiation flasks (which was clean and dried prior using water and kimtech wipes)	AE-588-10	/
		The flask containing AE-588-10 was equipped with a Equipment - Robust probe for liquid O2 measurement and placed in the Equipment - Advanced irradiation setup V1.0 I	/	/
	10:23	The degassing of AE-588-10 was started according to Protocol - Degassing and O2 Measurement with sensor cap with air stable substances	/	degas 2.jpg
	10:24	The oxygen logging was started	/	/

	10:29	An UV/Vis of AE-588-300-2 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I)	AE-588-300-2	AE-588-300-2.jpg
	10:31	An UV/Vis of AE-588-10-1 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I)	AE-588-10-1	AE-588-10-1.jpg
	10:33	The degassing was stopped	/	/
	10:34	The logging was stopped	/	/
	10:34	The oxygen logging was started	/	/
	10:42	The irradiation was started after 8:25	/	irrad 2.jpg
	11:02	The irradiation was stopped after 28:25	/	/
	11:08	The logging was stopped	no significant change	after irrad 2.jpg
	11:11	From AE-588-10 0.4 mL were transferred into another vial and 0.8 mL water was added	AE-588-10-2	/
	11:18	An UV/Vis of AE-588-10-2 was measured (Protocol - UV/vis measurements using Avantes spectrometer, Lab E208, CEEC I), for that 0.25 mL was diluted with 1.25 mL Milli-Q water	AE-588-10-2	/

Analysis

Used calibration:

- FireSting: [Prep work - AE-401: Liquid-phase calibration of Trace Range Robust Oxygen Probe I](#)

Date	Time	Sample name	Analysis method	Analytical device	Solvent	Raw Data	Python script	Processed Data	Comparative Data	Interpretation

13.10	09:23	AE-588-300-degas	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-588-300-degas.txt	/	AE-588-300-degas.png	HTE - AE-579: Irradiation of [Ru(bpy)3] after irradiation of 300 µM Ru(bpy)	worked
13.10	09:32	AE-588-300-irrad	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-588-300-irrad.txt	/	AE-588-300-irrad.png		Increase no O2 observed
13.10	10:24	AE-588-10-degas	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-588-10-degas.txt	/	AE-588-10-degas.png		worked
13.10	10:34	AE-588-10-irrad	Optical O2 detection	Equipment - Firesting Fiber-Optic Oxygen Meter 2 Channel	H2O	AE-588-10-irrad.txt	/	AE-588-10-irrad.png		increase till 20 µM, then decrease to 17 µM
13.10	09:59	AE-588-300-1	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-588-300-1.ABS8 AE-588-300-1_7420287SP.TXT	UV_Vis_multiple_spectra.py	AE-588-UVVis.png		df = 3 as expected, Ru(II)
13.10	10:29	AE-588-300-2	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-588-300-2.ABS8 AE-588-300-2_7420287SP.TXT	UV_Vis_multiple_spectra.py	AE-588-UVVis.png		df = 3 as expected, dimeric / polymeric species
13.10	10:31	AE-588-10-1	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-588-10-1.ABS8 AE-588-10-1_7420287SP.TXT	UV_Vis_multiple_spectra.py	AE-588-UVVis.png		df = 3 Small increase in orange dimer --> likely shoulder from Ru(II)
13.10	11:19	AE-588-10-2	UV/Vis	Equipment - Avantes UV/vis spectrophotometer	H2O	AE-588-10-2.ABS8 AE-588-10-2_7420287SP.TXT	UV_Vis_multiple_spectra.py	AE-588-UVVis.png		df = 3 significant increase in signal for dimeric/polymeric species

Results

Reproduction worked.

Bit more O2 formation (approx. 15 µM, instead of approx. 10 µM)

Linked experiments

HTE - AE-563: Irradiation of [Ru(bpy)3] after irradiation of 100 µM Ru(bpy)

HTE - AE-566: Irradiation of [Ru(bpy)3] after irradiation of 300 µM Ru(bpy) I

HTE - AE-576: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O I

HTE - AE-579: Irradiation of [Ru(bpy)3] after irradiation of 300 µM Ru(bpy)

HTE - AE-589: Preparation of stock solutions for the irradiation of [Ru(bpy)3]Cl2 * 6 H2O

Prep work - AE-401: Liquid-phase calibration of Trace Range Robust Oxygen Probe I

Prep work - AE-425: Liquid-phase calibration of Trace Range Robust Oxygen Probe under static conditions

Linked resources

Equipment - [VWR® VV3, Vortex Mixer](#)

Equipment - [Firesting Fiber-Optic Oxygen Meter 2 Channel \(Firesting 2\)](#)

Equipment - [Robust probe for liquid O2 measurment](#)

Equipment - [Advanced irradiation setup V1.0 I](#)

Equipment - [Irradiation setup 2 \(CEEC II, E004\)](#)

Light Source - [UHP LED 470 nm-1](#)

Power Meter - [843-R-USB + 919P-020-12](#)

Power Sources - [BLS-13000-1E](#)

Protocol - [Degassing and O2 Measurement with sensor cap with air stable substances](#)

Protocol - [UV/vis measurements using Avantes spectrometer, Lab 208, CEEC I](#)

Attached files

after irrad 2.jpg

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irrad 2.jpg

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AE-588-10-1.jpg

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degas 2.jpg

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with new Ru.jpg

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after irrad 1 bad quality.jpg

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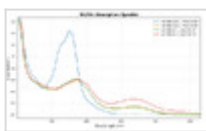
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UV_Vis_multiple_spectra.py

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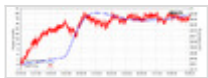
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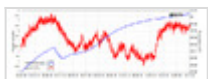


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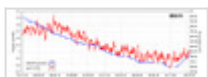
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AE-588-300-degas.png

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Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=3179>